

LEWISTOWN, MT, USA

WMO: 726776

Lat: 47.049N

Lon: 109.458W

Elev: 4145

StdP: 12.62

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24036

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-8.3	-21.6	1.9	-13.1	-14.4	2.9	-7.3	32.9	40.3	29.3	39.4	6.5	130	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.8	90.1	60.7	87.0	60.1	83.7	59.2	64.3	82.2	62.5	80.4	60.9	78.9	9.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.0	84.0	70.1	55.9	77.7	67.9	54.2	72.8	66.6	31.6	82.0	30.1	80.0	28.9	79.0	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.6	24.3	20.6	-23.0	95.8	7.0	2.6	-28.1	97.7	-32.2	99.2	-36.1	100.7	-41.2	102.6	
			-22.4	67.5	6.0	2.0	-26.7	68.9	-30.2	70.1	-33.6	71.3	-37.9	72.7	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	23.9	24.1	32.7	40.8	49.6	57.9	66.8	65.5	56.1	43.7	32.7	24.3
	DBStd	18.57	13.96	14.06	12.37	9.35	7.92	6.95	6.22	6.34	8.84	10.05	12.80	13.34
	HDD50	4092	809	725	540	297	104	13	0	1	39	240	526	798
	HDD65	8159	1273	1144	1001	725	478	230	55	73	286	661	970	1263
	CDD50	1650	0	1	4	22	93	251	522	481	224	45	7	0
	CDD65	239	0	0	0	0	1	17	112	88	20	1	0	0
	CDH74	4278	0	0	0	9	71	331	1793	1542	497	35	0	0
	CDH80	1576	0	0	0	0	7	87	724	592	160	6	0	0
Wind	WSAvg	9.2	10.3	9.6	9.8	9.7	9.2	8.5	7.5	7.5	8.1	9.3	10.2	10.2
Precipitation	PrecAvg	18.0	1.0	0.7	0.9	1.2	3.3	3.1	1.8	1.9	1.5	1.1	0.7	0.9
	PrecMax	28.1	2.8	2.0	2.6	2.7	9.2	6.5	4.8	5.6	3.9	2.4	1.5	2.4
	PrecMin	11.2	0.1	0.1	0.2	0.4	0.8	1.0	0.1	0.3	0.1	0.0	0.1	0.2
	PrecStd	4.6	0.7	0.4	0.6	0.7	2.0	1.6	1.2	1.5	1.1	0.7	0.4	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	57.3	66.8	74.9	80.6	87.9	94.9	94.2	89.5	78.8	67.6	54.3
		MCWB	40.5	42.1	46.2	51.2	56.8	62.4	62.8	61.1	58.8	53.3	48.0	41.2
	2%	DB	48.3	51.2	60.7	69.0	75.3	82.1	90.7	89.9	84.8	71.8	60.6	48.7
		MCWB	37.7	39.6	44.2	49.1	55.0	61.2	61.5	59.7	57.1	50.4	44.5	38.2
	5%	DB	44.5	46.1	54.7	63.4	71.2	77.5	87.6	86.4	80.2	65.9	55.0	44.9
		MCWB	36.2	36.7	41.6	46.9	53.4	59.3	61.2	59.3	55.6	48.6	41.9	36.2
	10%	DB	41.3	41.8	49.7	57.7	66.1	73.3	84.1	83.0	74.7	60.5	50.4	41.5
		MCWB	34.4	34.3	39.0	43.8	51.0	57.3	60.3	58.5	53.9	45.9	39.8	34.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.7	43.5	47.6	53.3	59.4	66.2	67.2	64.5	61.9	55.1	49.5	42.3
		MCDB	52.5	55.6	64.1	71.0	74.6	81.7	85.1	84.0	79.8	74.7	65.9	52.7
	2%	WB	38.7	40.2	45.0	50.0	56.6	63.1	65.1	62.4	59.1	51.7	45.3	39.0
		MCDB	47.3	50.1	59.4	66.4	71.7	77.8	83.6	82.6	79.8	67.7	58.9	47.6
	5%	WB	36.6	37.2	42.2	47.4	54.4	60.6	63.3	60.8	57.0	49.3	42.6	36.6
		MCDB	43.7	45.4	53.8	62.2	68.4	74.9	81.3	80.9	76.5	63.2	53.9	44.0
	10%	WB	34.7	34.8	39.4	44.9	52.0	58.5	61.7	59.5	54.7	46.9	40.1	34.6
		MCDB	41.0	41.6	48.9	57.0	64.1	71.5	79.9	78.8	72.2	59.6	49.9	41.2
Mean Daily Temperature Range	5% DB	MDBR	18.6	18.9	20.5	21.9	22.4	23.5	28.8	29.0	26.4	22.9	20.2	18.8
		MCDBR	20.9	22.4	27.6	32.1	31.3	31.2	34.3	35.1	35.4	32.4	27.2	20.8
		MCWBR	13.6	14.2	15.3	16.5	15.6	15.0	13.1	13.3	14.8	16.1	15.4	13.5
	5% WB	MCDBR	19.5	21.2	26.3	29.7	28.2	28.3	30.1	30.4	31.9	29.4	25.6	19.8
		MCWBR	13.3	13.9	15.2	15.8	14.6	14.4	13.5	12.5	14.1	15.4	15.0	13.4
Clear-Sky Solar Irradiance	taub	0.245	0.251	0.266	0.304	0.322	0.326	0.332	0.355	0.310	0.281	0.252	0.241	
	taud	2.524	2.545	2.577	2.462	2.465	2.493	2.495	2.389	2.506	2.572	2.567	2.518	
	Ebn at Noon	279	300	309	303	299	297	293	282	288	283	275	268	
	Edh at Noon	19	23	26	33	34	33	33	35	28	23	18	17	
All-Sky Solar Radiation	RadAvg	478	786	1164	1520	1811	2105	2262	1884	1380	880	542	377	
	RadStd	30	57	77	77	169	136	103	107	101	63	43	30	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page